

AI-Driven Recommendations for Short-Form Content

EDINBURGH AI





Welcome to the first workshop of
the semester!

What we'll be covering today

- ▶ What do Meta actually want from Instagram Reels in 2026 from a business perspective?
- ▶ The basics of modern short-form content recommendation (taking a focus on Instagram reels specifically)
- ▶ Implementing your own AI model for content recommendation.
- ▶ Doomscrolling



But first:
Why do reels even exist?

What Reels Are Trying to Achieve (Business Perspective)

Maximize Meaningful User Retention (Not Just Mindless Scrolling)

- ▶ *What?* Keep users coming back because they're genuinely getting value from the content, not just trapping their attention for as long as possible in one sitting.
- ▶ *Why?* Retention isn't the end goal for Meta, it's the mechanism. A user who sticks around longer sees more ads, generates more behavioural data to refine targeting, and is more likely to engage rather than tune out or install an ad blocker mentally. Meaningful retention also compounds: happy users stay for years, invite friends, and don't churn the way users who feel manipulated eventually do. So it's less "retention for its own sake" and more "retention as the multiplier on everything else below."

Show Adverts

- ▶ *What?* Insert ads into the Reels feed at intervals, targeted based on what the algorithm knows about the user.
- ▶ *Why?* Ads are the primary revenue driver for most short-form platforms. More time-in-app and better targeting generally means more ad impressions and higher ad revenue.

Gather User Data

- ▶ *What?* Track what users watch, skip, like, rewatch, and how long they pause on each piece of content.
- ▶ *Why?* This data trains the recommendation algorithm to show better content and also makes ad targeting more precise, which increases what advertisers are willing to pay.



So, what happens when Instagram
decides which reels to show you?

It's a Multi-Stage Process

If you tried to apply one massive AI model to every Reel in the world simultaneously, you'd be waiting a long time for anything to load.

To put this in perspective, Meta users view over 200 billion Reels a day, and new ones are being uploaded constantly. No single model checking every video against every other video in real time could keep up.

How do we solve this?

The idea is that we use smaller (but not as accurate) models to narrow down the candidate pool, and once we've narrowed down the search space we can use a bigger and better model

The General Pipeline

First off, it must be stated that while meta make some of the information on this public, they obviously keep many elements of it private. This is really just a general overview of our best guess.

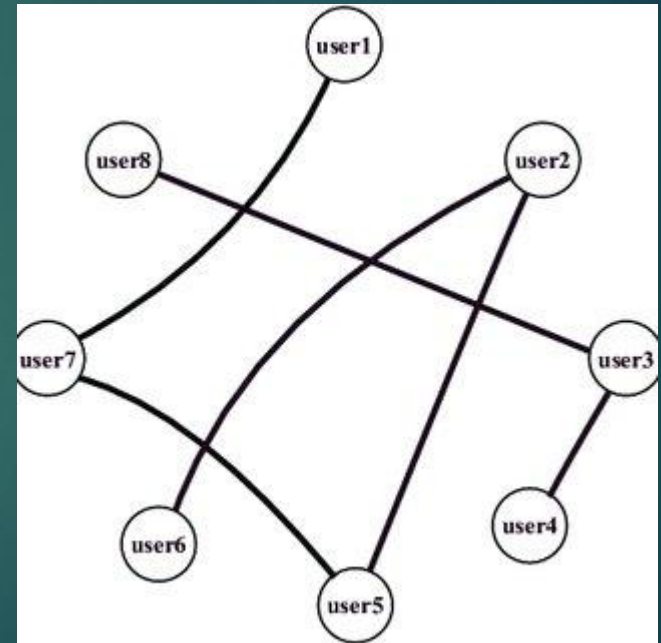


Stage 1: Lighter models

Meta uses a combination of models working in parallel here.

The 2 most interesting models at this stage are:

- ▶ **Two tower model**
- ▶ **User/Reel graph**



Stage 2: The heavier classifier

How Meta Does This: HSTU

Meta uses a model called **HSTU** (Hierarchical Sequential Transduction Units) to tackle this.

The core idea: treat a user's scrolling session as a single sequence, alternating between what they watched and what they did about it.

Video → Action → Video → Action → Video → Action → ...

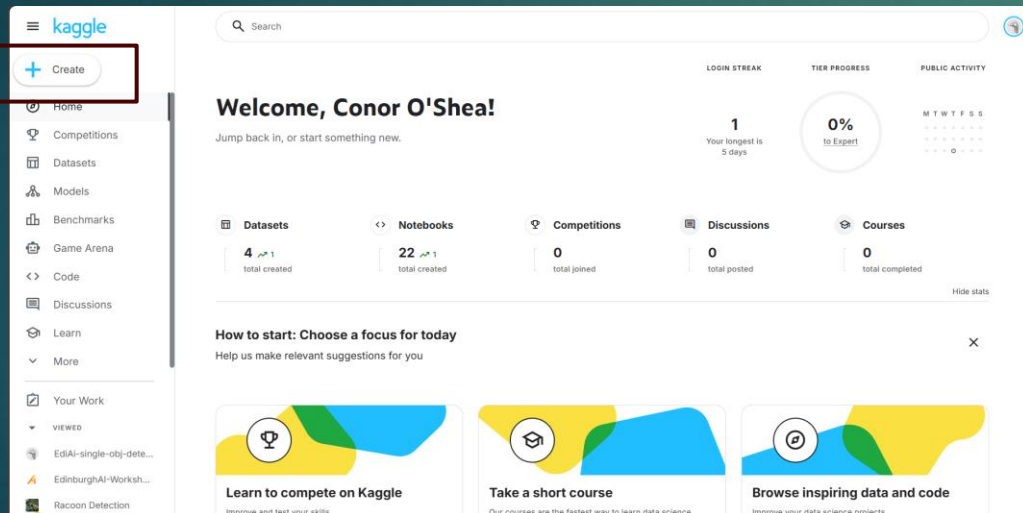
The model then looks at this sequence and predicts what action the user is likely to take on each new candidate reel, for example, whether they'll watch it fully, like it, or scroll past.



Let's have a go at building this!

Kaggle Setup

1. Go to: <https://www.kaggle.com/>
2. Create an account (if you don't have one already)
3. Click the create button in the top left and select “Import Notebook” from the dropdown menu.



5. Select “Link”
6. Enter one of the links on the following slide depending on which track you wish to follow

Workshop Notebook Tracks

Beginner Track:

- Recommended for those with little to no machine learning experience.
- Accessible to those who've never programmed before (no programming is required)
- Link: [workshops/PracticalWorkshops/Instagram-Reels-Recommendation/BEGINNER-ediai-reels-workshop.ipynb at main · EdinburghAI/workshops](#)

Advanced Track:

- Recommended to those with machine learning experience in PyTorch (ideally should know broadly how to implement a vanilla transformer)
- Quite a bit harder
- Link to notebook to complete: [workshops/PracticalWorkshops/Instagram-Reels-Recommendation/UNSOLVED-advanced-ediai-reels-workshop.ipynb at main · EdinburghAI/workshops](#)
- Link to notebook solutions: [workshops/PracticalWorkshops/Instagram-Reels-Recommendation/SOLVED-advanced-ediai-reels-workshop.ipynb at main · EdinburghAI/workshops](#)